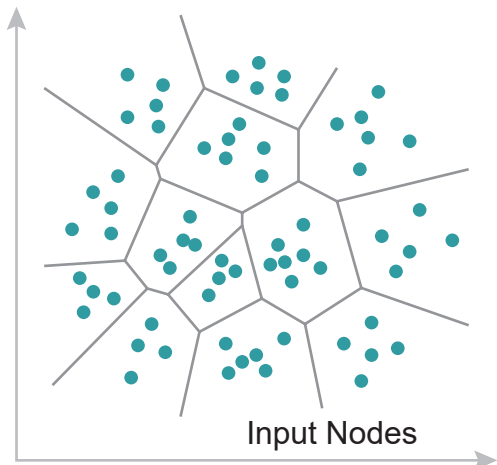


Geometric Pooling & Feature Aggregation

Before $\longrightarrow$

Input Space (Level ℓ)



Input Nodes

Geometric Clustering

$$\mathbf{X}^{(l+1)} = \mathbf{S}^T \mathbf{X}^{(l)}$$

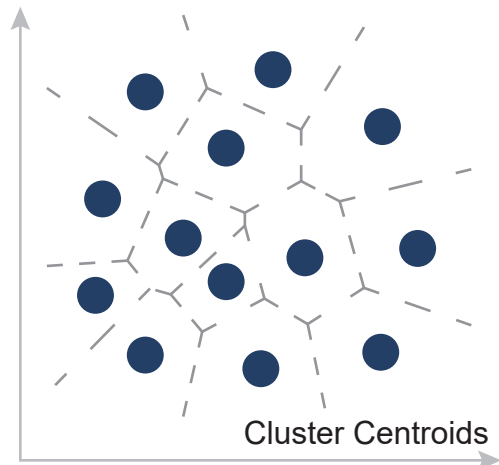


$$\begin{matrix} \mathbf{S} & \mathbf{X}^{(l)} & \mathbf{X}^{(l+1)} \\ \begin{array}{c} \begin{array}{|c|c|c|c|} \hline \text{teal} & \text{gray} & \text{gray} & \text{gray} \\ \hline \text{gray} & \text{teal} & \text{gray} & \text{gray} \\ \hline \text{gray} & \text{gray} & \text{teal} & \text{gray} \\ \hline \text{gray} & \text{gray} & \text{gray} & \text{teal} \\ \hline \text{teal} & \text{gray} & \text{gray} & \text{gray} \\ \hline \end{array} & \times & \begin{array}{|c|c|c|c|c|} \hline \text{teal} & \text{teal} & \text{teal} & \text{teal} & \text{teal} \\ \hline \text{teal} & \text{teal} & \text{teal} & \text{teal} & \text{teal} \\ \hline \text{teal} & \text{teal} & \text{teal} & \text{teal} & \text{teal} \\ \hline \text{teal} & \text{teal} & \text{teal} & \text{teal} & \text{teal} \\ \hline \text{teal} & \text{teal} & \text{teal} & \text{teal} & \text{teal} \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline \text{dark teal} & \text{dark teal} & \text{dark teal} & \text{dark teal} \\ \hline \text{dark teal} & \text{dark teal} & \text{dark teal} & \text{dark teal} \\ \hline \text{dark teal} & \text{dark teal} & \text{dark teal} & \text{dark teal} \\ \hline \text{dark teal} & \text{dark teal} & \text{dark teal} & \text{dark teal} \\ \hline \end{array} \\ N \times K & & N \times D & & K \times D \end{matrix}$$

Column-Normalized
Aggregation

$\longrightarrow$ After

Output Space (Level $\ell+1$)



Cluster Centroids

Dimensionality Reduction